



EXECUTIVE SUMMARY

Location-bound incident evidence for insurance

An enterprise verification platform that helps insurers determine whether claim evidence is consistent with the asserted place, time, people and vehicle.

DOCUMENT

IncidentProof operating model

VERSION

July 2026

MARKET

Indian insurance enterprises

STATUS

Product and pilot blueprint

01 / THE COMPANY

The proposition

Merklit is evidence infrastructure for insurers. It collects claim-related facts through web, mobile, surveyor and telematics channels, tests their consistency, and returns a verifiable incident-evidence result.

Merklit does not claim that a phone alone proves an accident. It proves individual facts and combines independent facts to support, contradict or qualify the incident narrative.

The insurer's question

Is the submitted evidence consistent with the claimant, vehicle, place and time described in the claim?

Merklit's answer

A structured set of evidence assertions, reason codes, integrity proofs and an auditable report for decision support.

Primary value**01****REDUCE AMBIGUITY**

Separate verified observations from claimant statements and unavailable evidence.

02**STRENGTHEN PROVENANCE**

Bind fresh media, time, location, actor and device evidence into one tamper-evident record.

03**SUPPORT REVIEW**

Route contradictions and weak evidence to human review instead of making automatic claim decisions.

RECOMMENDED POSITIONING

Location-bound incident evidence infrastructure for insurers. The commercial story is faster, more consistent claims review. Blockchain is an integrity layer, not the headline.

02 / WHAT IS VERIFIED

Five distinct assertions

Merklit reports each assertion independently. This prevents a weak signal from being converted into an unjustified statement that an incident definitely occurred.

ASSERTION	DEFENSIBLE MEANING	TYPICAL EVIDENCE
Policyholder presence	The authenticated policyholder's enrolled device produced acceptable evidence from the claimed region and time.	Login or OTP, device binding, location predicate, challenge and attestation.
Media captured here	A photograph or video was freshly captured inside the controlled Merklit workflow at the reported location and remained unchanged.	Camera-only capture, content hash, location, time and signed manifest.
Surveyor visited	The assigned and authenticated field representative produced visit evidence at the vehicle or property.	Roster identity, assignment, arrival proof, guided inspection and departure.
Vehicle event	A vehicle-linked source reported a collision-like event at a particular time and location.	OEM, fleet, OBD or telematics event correlated to policy or VIN.
Incident corroborated	Independent evidence sources are mutually consistent with the submitted incident account.	Presence, fresh media, vehicle event, surveyor evidence and external records.

Evidence verdicts

SUPPORTED	PARTIALLY SUPPORTED	INCONSISTENT	INSUFFICIENT
Required evidence is present and materially consistent.	Some assertions are supported, while required corroboration is absent.	One or more observations conflict with the claim narrative.	Evidence is unavailable, too weak or outside policy thresholds.

Decision boundary: Merklit recommends review; the insurer makes the claim decision. No `INCONSISTENT` or `INSUFFICIENT` result should automatically reject a claim.

One platform, multiple assurance levels

Clients may activate any combination of channels. The interface may be branded and customised, but the report must preserve the actual assurance level of the capture channel.

CHANNEL	AVAILABLE CAPABILITY	BEST USE	CANNOT CLAIM
ClaimLink	SMS or WhatsApp session, foreground location, live camera, server challenge and immediate hashing.	Fast policyholder evidence collection without installation.	Past presence, continuous tracking or hardware-grade device integrity.
Branded claim site	Full managed claim form, guided capture, document upload and insurer branding.	Insurers needing a complete digital claim journey.	Native attestation or reliable background sensing.
Web SDK	Embeddable JavaScript components, API session and configurable capture steps.	Existing insurer websites and portals.	Native-only sensor or secure-element capabilities.
Native SDK	Controlled camera, app/device attestation, hardware keys, sensor burst, offline chain and trip mode.	High-assurance evidence inside an insurer application.	Vehicle impact without a reliable event source.
Surveyor app/SDK	Native evidence plus employee identity, assignment, check-in, inspection and departure.	Field inspection and attendance disputes.	Policyholder presence at the original incident.
Telematics connector	Vehicle-linked time, location, impact and motion event ingestion.	Automatic event corroboration.	Human presence or media authenticity.
Evidence import API	Police, garage, towing, dashcam and partner evidence with source attribution.	One consolidated claim evidence record.	Pre-receipt integrity unless the source signed the evidence.

Key rule: a branded website remains a web-assurance channel. Visual customisation must never be presented as technical assurance.

Configurable incident triggers

MANUAL INCIDENT MODE

The user presses a button at the scene. Merklit runs a short location and sensor session, collects guided media, and seals the evidence. This is the recommended policyholder starting point.

CLAIM-FORM CAPTURE

Merklit binds live evidence to the claim form. At the scene, it supports the incident location. If completed later, it verifies only the submission or vehicle-inspection location.

POSSIBLE CRASH PROMPT

A native app may use permitted motion and trip signals to prompt evidence collection. It remains a trigger because hard braking, a dropped phone and OS restrictions can create false or missed events.

TELEMATICS TRIGGER

A signed OEM, fleet or OBD event opens a case and requests policyholder evidence. This is the strongest automatic motor-incident trigger.

TRIP EVIDENCE MODE

An explicitly enabled trip session creates adaptive encrypted commitments. A rolling local buffer may be disclosed after an incident under the configured consent and retention policy.

SURVEYOR ASSIGNMENT

The insurer assigns a visit. The authenticated surveyor creates arrival, inspection and departure proofs against the claim and assigned asset.

CONTINUOUS LOCATION IS AN OPTIONAL MODE, NOT A UNIVERSAL PROMISE

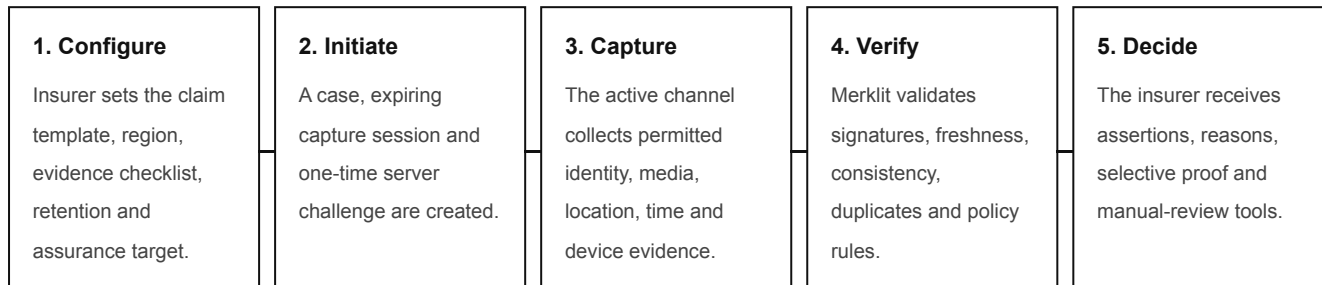
Mobile operating systems restrict background location and motion processing. Merklit should use user-visible incident or trip sessions, adaptive sampling and explicit permission rather than promise an exact five-minute snapshot on every device.

Recommended trigger hierarchy

PRIORITY	MODE	REASON
1	Manual incident + guided capture	Technically achievable, understandable and immediately useful.
2	Surveyor assignment	Controlled workforce and measurable insurer value.
3	Telematics	High-value corroboration once commercial partnerships exist.
4	Crash heuristics and trip mode	Useful advanced modules after device and battery testing.

05 / OPERATING MODEL

End-to-end claim evidence flow



- 1. Case creation:** the insurer sends claim identifiers, actor roles, expected region, allowed channels and the required capture template to `POST /v1/cases`.
- 2. Fresh session:** Merklit returns an expiring session and nonce so old evidence cannot be replayed into a new claim.
- 3. Consent and binding:** the actor receives a purpose-specific notice and authenticates through the insurer, OTP, enterprise SSO or surveyor identity.
- 4. Controlled capture:** evidence is captured through the browser or native SDK. Each permitted observation is labelled with source, availability and accuracy.
- 5. Immediate integrity:** media and evidence components are hashed as close to capture as the channel allows. Native channels sign with a hardware-backed key where supported.
- 6. Secure ingest:** the API validates the nonce, schema, signatures, attestation and replay state before accepting encrypted evidence.
- 7. Consistency engine:** the system compares location, time, actor, media, movement, device risk, telematics and third-party records.
- 8. Proof creation:** item roots form a claim root; claim roots enter a tenant batch for trusted timestamping and public anchoring.
- 9. Review and disclosure:** the portal shows the verdict for each assertion. Raw or exact-location evidence remains restricted unless an authorised escalation is approved.
- 10. Export:** APIs, webhooks and evidence packages deliver the result to the insurer's claims, fraud, legal and audit workflows.

Default response: the insurer receives whether the claimed region and evidence requirements were supported, not an unrestricted dump of personal location history.

Enterprise architecture

EXPERIENCE	ClaimLink, managed claim site, JavaScript SDK, iOS/Android SDK, surveyor app and insurer portal.
API EDGE	API gateway, tenant authentication, rate limits, session service, nonce/replay protection and webhooks.
CAPTURE CONTROL	Templates, consent, actor binding, device capability detection, offline queue and media ingestion.
VERIFICATION	Signature and attestation validation, location policy, media provenance, consistency rules, duplicate detection and reason codes.
EVIDENCE	Encrypted object vault, structured metadata store, append-only audit trail, retention and legal hold.
CRYPTOGRAPHY	Per-evidence hashes, hardware or service signatures, claim Merkle tree, tenant batch root, CCA timestamp and Base anchor.
ENTERPRISE	SSO/SCIM, KMS or BYOK, role-based access, tenant isolation, monitoring, data export and dedicated deployment.

Proof structure

EVIDENCE LEAVES Media hash, location commitment, trusted-time evidence, actor/session, attestation, capture policy, ruleset and quality output.	CLAIM ROOT All evidence-item roots and review events for a single claim are combined into a Merkle root.	BATCH ANCHOR Multiple tenant-safe claim commitments become a batch root for timestamping and Base publication.
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Data protection model

- A random AES-256-GCM data key encrypts each evidence object.
- Keys are wrapped by a tenant-specific KMS key; enterprise clients may use BYOK or dedicated infrastructure.
- PII, raw media, policy numbers and coordinates never enter the blockchain.
- Exact-location and complete-report access require a policy-controlled, audited escalation.
- Control-plane configuration is separated from tenant evidence data and cryptographic keys.

Practical product capabilities

INCIDENT CAPTURE

- Guided photo and video
- Camera-only capture
- Configurable claim checklists
- Offline encrypted queue
- Recapture and quality guidance
- Regional-language workflows

LOCATION AND TIME

- Region, polygon and radius checks
- Accuracy and freshness thresholds
- Altitude and motion where available
- Clock anomaly and monotonic time
- Coverage-gap reporting
- CCA trusted timestamp

DEVICE AND IDENTITY

- Insurer login, OTP and SSO
- Surveyor roster and assignment
- App Attest and Play Integrity
- Hardware-backed signing keys
- Root, jailbreak and emulator signals
- Replay and duplicate-session protection

MEDIA PROVENANCE

- Immediate SHA-256 content hash
- EXIF consistency checks
- Duplicate and reused-media search
- Blur, darkness and glare detection
- Manipulation indicators
- Optional C2PA Content Credential

CORROBORATION

- Telematics, OEM and OBD events
- Surveyor and garage evidence
- Towing, police and dashcam import
- Cross-party consistency
- External event-source snapshots
- Vehicle and policy association

REVIEW AND OPERATIONS

- Claim timeline and evidence map
- Assertion-level verdicts
- Reason codes and missing signals
- Manual-review queue
- Assignments, overrides and approvals
- SLA and performance analytics

INTEGRATION

- REST/OpenAPI and webhooks
- JavaScript, Swift and Kotlin SDKs
- React Native wrapper
- Sandbox and test harness
- Claims-system adapters
- SSO, SCIM and storage connectors

PRIVACY AND LEGAL

- Purpose-specific consent
- Selective regional disclosure
- Role-based evidence access
- Retention, deletion and legal hold
- Section 63 export workflow
- Independent QR or hash verification

Capability principle: unavailable device signals must be reported as unavailable. They must never be silently replaced, inferred or represented as captured facts.

Tamper evidence, correctly framed

BLOCKCHAIN PROVES

- The exact evidence commitment existed by a recorded block.
- A later change would produce a different hash.
- The public anchor can be checked independently of Merklit.

BLOCKCHAIN DOES NOT PROVE

- That GPS or a sensor input was truthful.
- That the claimant personally held the device.
- That an accident occurred merely because a hash exists.

Recommended integrity sequence

1. Hash evidence components at capture and bind them to the fresh claim session.
2. Sign the evidence manifest with the available hardware-backed or Merklit service key.
3. Create the claim Merkle root and issue an immediate signed receipt.
4. Obtain a timestamp from an Indian licensed Certifying Authority.
5. Batch claim commitments and publish only the batch root to Base.
6. Preserve the transaction, inclusion path, verifier version and certificate status for long-term validation.

Indian legal package

The export should include the evidence manifest, hashes, signatures, capture source, relevant device particulars, trusted timestamp, blockchain inclusion proof, audit trail, system and ruleset versions, and a configurable Bharatiya Sakshya Adhinyam Section 63 certificate workflow.

Legal position: describe the output as a court-supporting electronic evidence package, not guaranteed admissibility. Admissibility and evidentiary weight remain subject to proper procedure, facts and legal review.

Controlled full-report escalation

The default response may disclose only whether the claimed region and evidence requirements were supported. A complete report should require an insurer-authorised request, stated purpose, maker-checker approval, least-privilege decryption and a permanent audit entry.

09 / PRODUCT PACKAGING

Productised core, configurable delivery

Approximately 80% of the platform should remain common across clients. Branding, rules, workflows, connectors and deployment policy provide the client-specific 20%.

MODULE	OFFER	COMMERCIAL UNIT
ClaimLink Verify	Hosted customer capture and current-location evidence without app installation.	Per completed evidence session.
Embedded Incident SDK	Native or web components inside the insurer's existing digital channels.	Platform fee plus proof volume.
Surveyor Proof	Assignment, attendance, inspection capture and evidence audit.	Per surveyor or completed visit.
Telematics Corroboration	Vehicle-event connectors and claim correlation.	Connector fee plus event volume.
Evidence Import	Partner, garage, police, towing and existing evidence ingestion.	API volume and connector fee.
Legal Proof Pack	Timestamp, blockchain proof, verification bundle and Section 63 workflow.	Per generated package.
Enterprise Deployment	Dedicated tenant, BYOK, custom retention, private connectivity and enhanced SLA.	Annual enterprise contract.

Commercial structure

IMPLEMENTATION

One-time discovery, configuration, security review, SDK integration and custom connector fees.

PLATFORM

Annual minimum commitment covering tenant, portal, support, security and core APIs.

USAGE

Volume pricing by completed proof, visit, event, legal export or agreed claim unit.

Custom websites and apps: offer them as professional services built over the shared platform. Do not maintain separate product forks for individual insurers.

10 / PILOT AND ROADMAP

Start with motor claims

The strongest initial wedge is motor own-damage first notification and surveyor inspection: clear actors, visible evidence, existing insurer workflows and measurable review outcomes.

1

DESIGN

Select one insurer and one claim type. Map current decision points, evidence gaps and legal requirements.

2

SHADOW PILOT

Run ClaimLink and Surveyor Proof on 500–2,000 claims without affecting claim approval.

3

SCALE

Add native SDK, telematics and automated rules only after measurable pilot value.

Pilot success measures

OPERATIONAL	EVIDENCE QUALITY	TECHNICAL	COMMERCIAL
Claim cycle time Recapture rate Surveyor dispute rate Manual-review time	Completion rate Contradiction yield Duplicate-media yield False-flag review	SDK stability Battery impact Permission acceptance Proof availability	Insurer savings Conversion to paid use Expansion modules Integration effort

Roadmap

PHASE 1	Case API, ClaimLink, guided media, current-location result, claim root, review portal and signed export.
PHASE 2	Android/iOS SDK, device attestation, hardware signatures, surveyor workflow, CCA timestamp and Base anchoring.
PHASE 3	Telematics connectors, C2PA media credentials, duplicate intelligence, advanced rules and insurer-system adapters.
PHASE 4	Trip mode, validated crash prompting, independent verification tooling, dedicated deployments and additional insurance lines.

Build-versus-partner: Merklit should own the insurer workflow, evidence graph, reports and integration layer. It may initially license or partner for hardened location proof while its own anti-spoofing engine is independently tested.

Success depends on disciplined scope

RISK	FAILURE MODE	REQUIRED RESPONSE
Overclaiming truth	Marketing says the phone or blockchain proves an accident.	Use assertion-level wording and always disclose unavailable evidence.
Mobile restrictions	Background capture becomes unreliable or drains battery.	Prioritise active incident/trip sessions and publish a device capability matrix.
Spoof resistance	Single-source location is manipulated.	Use attestation, independent signals, freshness, telematics and adversarial testing.
Privacy	Continuous or exact-location collection exceeds the claim purpose.	Minimise collection, disclose regional results and enforce retention and access controls.
Custom-project trap	Every insurer receives a separate codebase.	Use tenant configuration, reusable connectors and documented extension points.
Enterprise sales	Long procurement without quantified value.	Start with a design partner, shadow pilot and agreed economic metrics.

Risk-adjusted probability

35%

SUSTAINABLE B2B COMPANY

Estimated probability of three or more paying insurers and repeatable annual revenue within three to five years.

45%

IMPROVED CASE

With a committed insurer design partner, narrow motor pilot, independent security review and productised delivery.

Indicative sub-ratings: technical MVP feasibility 80%; hardened adversarial location capability 60%; first paid insurer pilot 50%; large venture-scale outcome 12–15%. These are judgement-based estimates, not statistical forecasts.

The opportunity is credible if Merklit sells better evidence decisions, not blockchain. The first proof of the company is a paid insurer pilot with measurable claims value.

12 / IMPLEMENTATION PRINCIPLES

What should change from the original concept

DO

- Hash each evidence component at capture.
- Use region-based selective disclosure by default.
- Use per-tenant encryption and optional BYOK.
- Report signal availability and confidence.
- Use CCA time plus a public blockchain anchor.
- Keep the verifier independent and versioned.
- Send weak evidence to manual review.

DO NOT

- Promise exact five-minute background capture.
- Depend on IMEI or SEID access.
- Collect heart-rate data without a valid need.
- Use biometric templates; store only auth success.
- Include a duress wipe in the insurer product.
- Put PII, media or coordinates on-chain.
- Describe blockchain as proof the event was true.

Recommended integration surface

SURFACE	DELIVERY	PURPOSE
REST API	Versioned OpenAPI specification	Cases, actors, sessions, evidence, verdicts, reports and tenant configuration.
Webhooks	Signed, replay-protected events	Case status, evidence completion, review required, verdict and export ready.
Web SDK	npm package and ES module	ClaimLink components, capture, consent and current-location sessions.
Native SDK	Swift Package and Android library	Attestation, hardware signing, controlled capture, offline queue and permitted sensors.
Wrappers	React Native and supported hybrid bridges	Insurer applications that do not use fully native presentation layers.
Verifier	Web, CLI and documented proof format	Independent signature, hash, timestamp and blockchain inclusion validation.

Architecture principle: selective disclosure, encrypted custody and full-report access are separate concepts. Merklit can disclose a regional result by default while retaining an encrypted, access-controlled report for authorised review.

13 / EXTERNAL REFERENCES

Research basis

This executive summary is a product and technical blueprint, not legal advice. The following primary or product-owner sources informed the feasibility boundaries and recommended controls.

1. **Octet:** hardware-signed predicate-based location proof, on-device sensor fusion and offline verification.
octetproof.com
2. **Truepic:** trusted media capture, time/location verification and insurance inspection workflows.
truepic.com
3. **Apple Core Location:** location capabilities and background execution behaviour.
developer.apple.com
4. **Android Developers:** background location limits and permission guidance.
developer.android.com
5. **Google Play Integrity:** application, device and environment integrity verdicts.
developer.android.com/google/play/integrity
6. **Android privacy changes:** restrictions on non-resettable device identifiers such as IMEI and serial number.
developer.android.com/about/versions/10
7. **Controller of Certifying Authorities:** Indian time-stamping service and licensed CA framework.
cca.gov.in/timestamping.html
8. **India Code:** Bharatiya Sakshya Adhiniyam, 2023, Section 63, admissibility of electronic records.
indiacode.nic.in
9. **Base:** Ethereum Layer 2 rollup architecture and transaction data publication.
docs.base.org
10. **C2PA:** open technical standards for cryptographically verifiable media provenance and Content Credentials.
spec.c2pa.org
11. **MeitY:** Digital Personal Data Protection Act and Rules documentation relevant to personal-location processing.
meity.gov.in
12. **IRDAI:** insurance regulation, outsourcing, governance and claims-related circulars and guidance.
irdai.gov.in

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